

Hemoglobinopathies & Thalassemias

Giri Raval • designed to follow alongside the 40-slide lecture deck

Official Week 7 LOs

Slide-order companion

ScholarRx links

Faculty companion details

Current 2026 updates separated

Factoid sheet companion

Overview: What the Official Week 7 Learning Objectives Want From You

LO 1. State the composition of normal hemoglobin at various stages of development and explain the oxygen-binding properties of fetal and adult hemoglobin.

LO 2. Define hemoglobinopathy, list the qualitative and quantitative types of hemoglobinopathies, and explain the genetic basis underlying these categories.

LO 3. Interpret hemoglobin composition analyses and use them to differentiate the major hemoglobinopathies.

LO 4. Describe the genetic and hematologic findings and clinical characteristics of the main structural hemoglobin variants: HbS, HbC, and HbE.

LO 5. Describe the genetic, hematologic, and clinical differences among alpha- and beta-thalassemias.

SCHOLARRX BRICKS PAIRED WITH THIS LECTURE

Hemoglobin Structure & Function: Molecular Design for Oxygen Transport

Sickle Cell Disease

The Genetic Basis of Select Thalassemias

These three Bricks are explicitly listed in the official Week 7 objective sheet.

LO 1

Slides 2-6: hemoglobin structure, normal Hb fractions, chromosome 16/11 globin genes, and developmental switching.

LO 2

Slides 7-10 + 28-31: qualitative structural variants versus quantitative thalassemias and their genetic mechanisms.

LO 3

Slide 3 + companion enrichment: normal fractions, then a dedicated composition-analysis framework below.

LO 4

Slides 11-27 + companion gap-fill: HbS is the slide focus; HbC/HbE are added from the uploaded companion handout because the slide deck does not develop them.

LO 5

Slides 28-40: alpha/beta thal genetics, chain imbalance, phenotypes, and treatment.

MENTAL MODEL BEFORE THE LECTURE STARTS

Hemoglobinopathy = either the globin protein is made wrong (qualitative) or the correct globin chain is made in the wrong amount (quantitative).

Structural variants change **what the chain is**. Thalassemias change **how much of the chain is produced**.

Slides 2-3: start with the molecule and the fractions

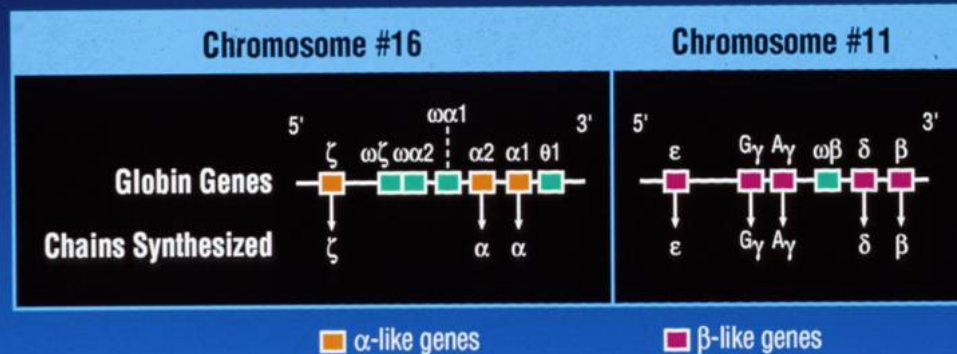
- Hemoglobin is a tetramer with **4 globin chains + 4 heme groups**.
- Normal adult hemoglobin on the lecture slide: **HbA (alpha2 beta2) 96-98%**, **HbA2 (alpha2 delta2) 2.5-3.5%**, and **HbF (alpha2 gamma2) <1%**.
- The slide's fetal column shows predominantly **HbF (80-85%)** with some HbA already present.
- Embryonic hemoglobins listed: **Gower-1, Gower-2, Portland-1**.

Hemoglobin	Chains	Adult abundance / role	Exam use
HbA	alpha2 beta2	96-98% adult	Loss/reduction points toward beta-chain disease or replacement by variant Hb.
HbA2	alpha2 delta2	2.5-3.5%	Elevation is a classic clue to beta-thalassemia trait.
HbF	alpha2 gamma2	<1% adult; dominant fetal Hb	High HbF protects against HbS polymerization and mitigates beta-thal chain imbalance.

WHAT THIS MEANS

Developmental switching matters because a mutation can be present from conception but remain clinically quiet while the affected globin chain is not yet the dominant one. Beta-chain disorders become clinically apparent only after gamma-globin/HbF falls and beta-globin production becomes important.

Chromosomal Organization of the Human Globin Genes



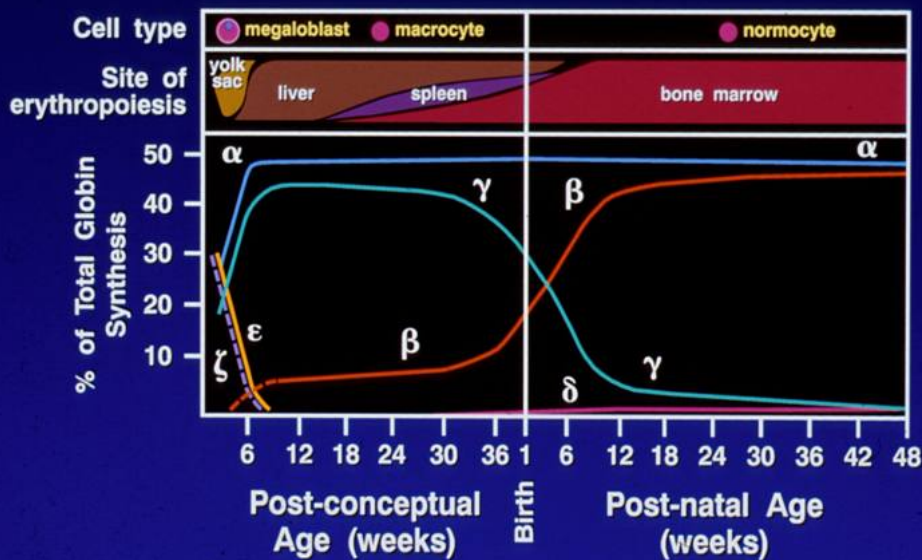
Lecture slide 4 - full slide retained. Chromosome 16 contains the alpha-like cluster; chromosome 11 contains the beta-like cluster. This genomic organization underlies the deletion-heavy genetics of alpha-thalassemia versus the mostly point/splice/nonsense mechanisms of beta-thalassemia.

NEED TO MEMORIZE

Alpha cluster = chromosome 16, four alpha genes total.

Beta cluster = chromosome 11, two beta genes total.

Hemoglobin Switching: Changes in Globin Chain Production and Sites of Hematopoiesis During Development



Lecture slide 6 - full slide retained. Watch both things on this figure: the site of erythropoiesis migrates toward bone marrow, while gamma production falls and beta production rises after birth.

DON'T MISS THE SOURCE DISCREPANCY

The lecture slide later states that the fetal-to-adult gamma-to-beta switch is near-complete at about **3 months**. The uploaded companion handout says beta-chain disease symptoms are delayed until about **6 months**. Keep the two source statements distinct: both are pointing to the same clinical idea - **HbF protects early infancy while the postnatal switch occurs over the first months of life.**

BOARD ENRICHMENT - WHY HbF HAS HIGHER OXYGEN AFFINITY

HbF binds 2,3-BPG less avidly than HbA, so its oxygen-dissociation curve is left-shifted. That allows fetal blood to extract oxygen from maternal blood. You do not need to turn this into a giant physiology detour; the useful association is **HbF = higher O₂ affinity + anti-sickling effect.**

LO 1 checkpoint

Without notes, can you write HbA/HbA2/HbF chain composition, approximate adult percentages, chromosome 16 versus 11, and explain why beta-chain diseases emerge after HbF wanes?

Slides 7-10

Qualitative versus quantitative hemoglobinopathies

LO 2 + LO 4

Slide 7 gives the classification you should use all week

- **Qualitative = Hb variants:** usually missense mutations that change globin structure. Examples: HbS, HbC, HbE.
- **Quantitative = thalassemias:** decreased or absent production of one or more globin chains.

	Qualitative / structural	Quantitative / thalassemia
Core problem	Abnormal amino-acid sequence / protein behavior	Too little or no production of a globin chain
Classic genetics	Missense substitution (HbS/C/E)	Alpha: often deletions; Beta: often promoter/splice/nonsense/frameshift mutations
Examples	HbS, HbC, HbE, unstable Hb, altered-affinity Hb	alpha-thal, beta-thal

Slides 8-10: why the location of the amino-acid substitution matters

- Surface substitutions can create abnormal intermolecular interactions - **HbS polymerization** is the prototype.
- Changes near the heme pocket can produce **unstable hemoglobin / Heinz bodies** or **methemoglobinemia**.
- Interchain-contact mutations can alter stability or O₂ affinity; high-affinity variants can drive compensatory erythrocytosis.

OFFICIAL LO GAP-FILL - HbC AND HbE ARE NOT DEVELOPED IN THE SLIDE DECK

The uploaded companion handout supplies the main recognition facts:

- **HbC:** beta6 Glu -> Lys; usually mild hemolytic anemia/splenomegaly; HbC crystals can be seen on smear.
- **HbE:** beta26 Glu -> Lys; common in Southeast Asia; behaves as both a structural variant and a mild beta-thalassemia because the mutation can alter splicing.

NEED TO MEMORIZE - POSITION 6

HbS: beta6 Glu -> Val.

HbC: beta6 Glu -> Lys.

Same position, completely different protein behavior.

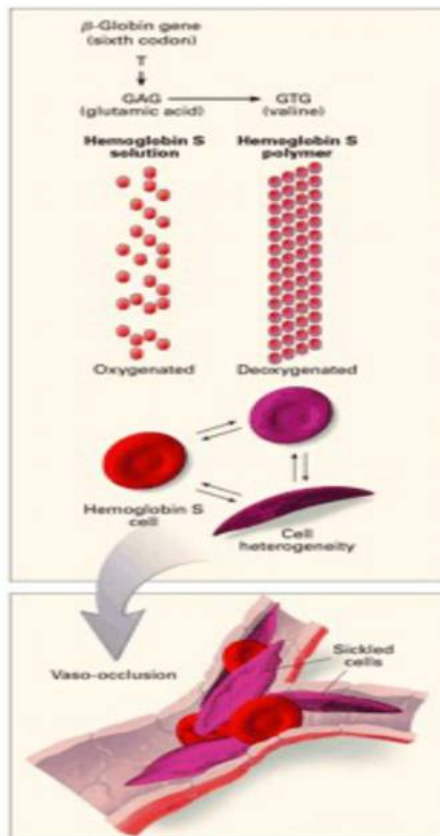
Slides 11-19

Sickle cell disease: mutation -> polymerization -> hemolysis + vaso-occlusion

LO
4

Slide 11: the molecular lesion

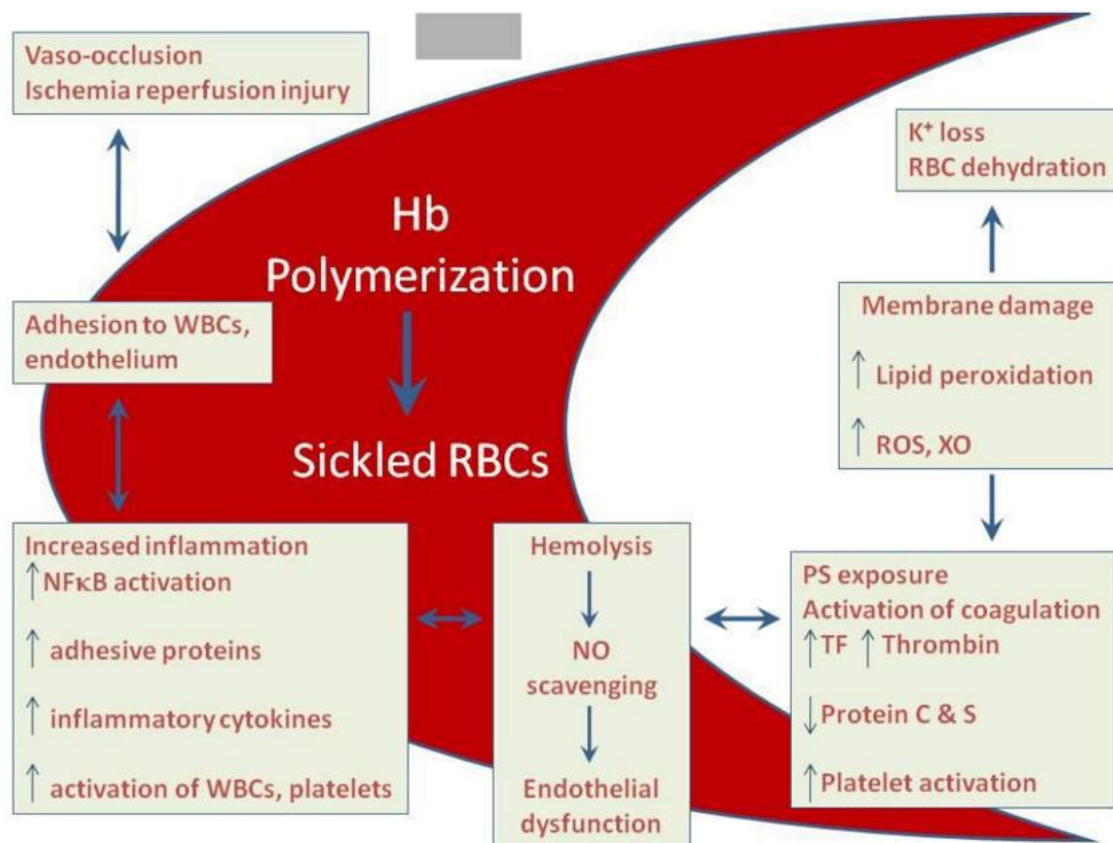
- Sickle cell anemia = homozygous HbS.
- **beta6 Glu -> Val**, with the lecture giving **GAG -> GTG**.
- The hydrophilic-to-hydrophobic substitution creates the abnormal interaction that permits deoxygenated HbS to polymerize.



Lecture slide 16 - full slide retained. The core disease mechanism: deoxygenated HbS forms polymers, distorting the RBC and creating the substrate for vaso-occlusion and membrane injury.

CAUSE -> EFFECT

Deoxygenation / higher intracellular HbS concentration -> polymerization -> rigid sickled RBC -> microvascular obstruction + repeated membrane damage -> ischemia and hemolysis.



Lecture slide 17 - full slide retained. This is the most important mechanistic slide in the SCD block. Polymerization is central, but inflammation, adhesion, dehydration, oxidative membrane damage, endothelial dysfunction, and coagulation feed back into the disease.

NEED TO MEMORIZE - WHAT PROMOTES POLYMERIZATION

Low O₂, acidosis, dehydration / increased MCHC.

HbF inhibits polymerization. HbA also dilutes/inhibits HbS polymer formation.

Slides 13-15: disease genotypes and phenotype severity

- **HbSS** = sickle cell anemia.
- **HbSC** = compound heterozygous S/C disease.
- **S beta⁰-thal** has no normal beta production from the thal allele and behaves more severely.
- **S beta⁺-thal** retains some beta production and generally has a milder phenotype.

DON'T MISS THE DENOMINATOR

“Sickle cell disease” is an umbrella category. **HbSS is one genotype within SCD.** The lecture deliberately lists SC and S/beta-thal variants because genotype changes the hemoglobin mixture and therefore disease severity.

Slides 18-19: complications are manifestations of two core processes

- **Vaso-occlusion / ischemia:** pain episodes, dactylitis, acute chest syndrome, stroke, avascular necrosis, priapism, nephropathy.
- **Chronic hemolysis:** anemia, bilirubin/gallstones, NO scavenging and endothelial dysfunction.
- **Functional asplenia:** infection/sepsis risk after progressive splenic injury.
- **Marrow stress:** aplastic crisis can sharply worsen anemia.

BOARD ENRICHMENT - CLASSIC INFECTIOUS CLUES

Functional asplenia increases risk from encapsulated organisms. In a vignette, **Salmonella osteomyelitis** is a classic association with SCD, while **parvovirus B19** points toward aplastic crisis.

LO 4 checkpoint - HbS

Can you go from **beta6 Glu->Val** to polymerization, then explain why dehydration/hypoxia worsen disease, why HbF helps, and sort major complications into vaso-occlusive versus hemolytic consequences?

Slides 20-27

HbF as a modifier, hydroxyurea, transfusion, transplant, and gene therapy

LO 1 + LO

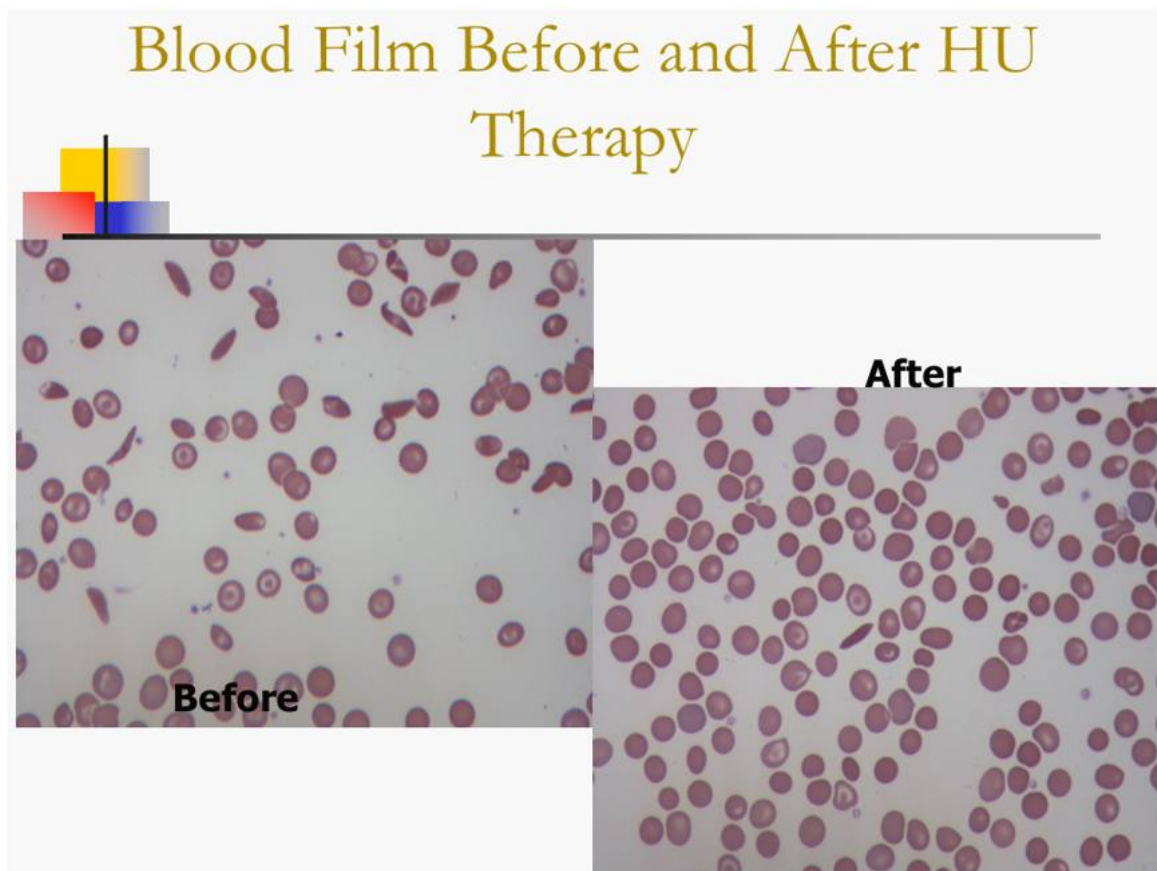
4

Slide 20 is the bridge between developmental biology and treatment

- High **HbF** is beneficial in both SCD and beta-thalassemia.
- In SCD, HbF reduces deoxy-HbS polymerization.
- In beta-thalassemia, gamma-chain production helps compensate for missing/reduced beta chains and reduces alpha/non-alpha imbalance.

NEED TO MEMORIZE

Hydroxyurea -> increases HbF -> fewer sickling events.



Lecture slide 22 - full slide retained. Use this slide qualitatively: hydroxyurea therapy is tied to HbF induction and changes in the sickling phenotype. Do not waste time memorizing individual cells in this image.

Slide 21: therapeutic buckets

- **Supportive / preventive care** and pain management.
- **Transfusion:** episodic or chronic depending on indication.
- **Disease-modifying:** hydroxyurea, L-glutamine, crizanlizumab; the slide also lists voxelotor and pyruvate kinase inhibitors.
- **Curative-intent approaches:** stem-cell transplantation and gene therapy.

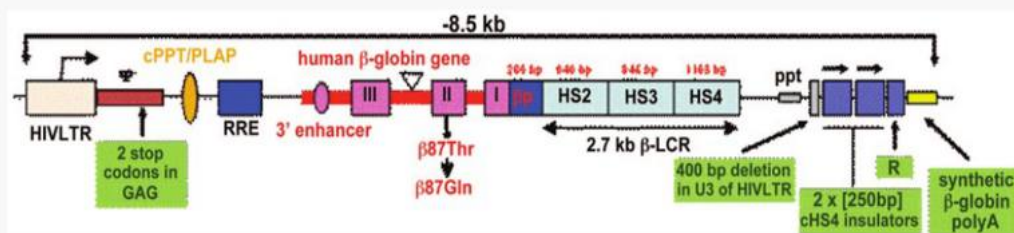
CURRENT 2026 UPDATE - IMPORTANT CORRECTION TO THE SLIDE

Voxelotor (Oxbryta) should not be treated as a current therapy. The FDA announced its voluntary market withdrawal in September 2024 because the emerging safety data changed the benefit-risk assessment. This is external/current information, not a correction silently inserted into the professor's slide.

FDA Oxbryta withdrawal notice

Sickle Cell Disease: Approaches to Gene Therapy

- Gene addition:
 - ❖ Add potent “anti-sickling” globin gene: β -T87Q
 - ❖ Add modified γ -globin with more potent anti-sickling effect
- Gene editing/correction
 - ❖ Erythroid specific knock down of BCL11a to increase Hb F
 - ❖ Correction of the sickle mutation (GAG→GTG) with CRISPR-Cas



Lecture slide 24 - full slide retained. The slide separates gene addition from gene editing. The common strategic theme is either create an anti-sickling beta-like globin or reactivate gamma-globin/HbF.

WHAT BCL11A EDITING IS ACTUALLY DOING

BCL11A normally suppresses gamma-globin after birth. Editing its erythroid regulatory element **releases that brake**, allowing HbF to rise. This does not directly repair the HbS nucleotide in the common Casgevy strategy.

CURRENT 2026 FDA UPDATE

The slide correctly highlights the 2023 approvals of **Casgevy (exa-cel)** and **Lyfgenia (lovo-cel)**. FDA labeling has continued to evolve: the current Casgevy page includes SCD with recurrent VOCs and transfusion-dependent beta-thalassemia, with newer 2026 approvals expanding age eligibility. Keep the mechanism distinction clear: Casgevy uses genome editing to increase HbF, while Lyfgenia adds an anti-sickling beta-globin gene.

FDA Casgevy • FDA Lyfgenia

Slides 28-32

Thalassemias: alpha vs beta genetics and chain imbalance

LO 2 + LO
5

Slides 28-31: classify the production defect

- **Alpha-thalassemia:** most commonly deletional. You have four alpha genes total, so phenotype follows the number and arrangement of deleted genes.
- **Beta-thalassemia:** usually point/splice/transcription/frameshift/nonsense mutations affecting the two beta genes.
- **beta+** = reduced beta-globin production.
- **beta0** = absent beta-globin production from that allele.

NEED TO MEMORIZE - GENETICS SHORTCUT

Alpha = deletions, chromosome 16, count genes.

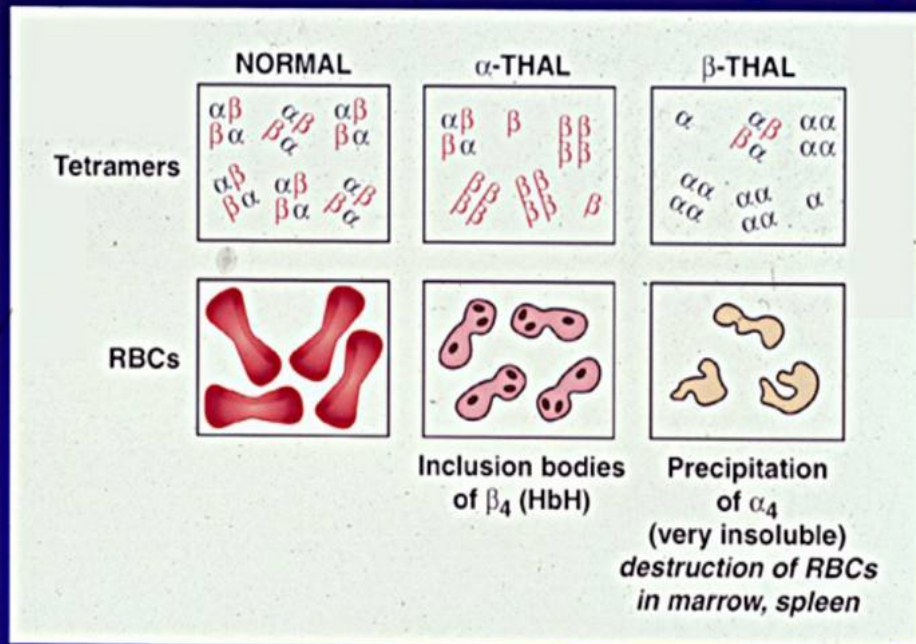
Beta = mutations, chromosome 11, beta+ versus beta0.

Alpha genes left	Genotype concept	Result	High-yield Hb
3	1 deletion	Silent carrier	Usually normal electrophoresis
2	2 deletions, cis or trans	Alpha-thal trait	Usually normal adult electrophoresis
1	3 deletions	HbH disease	beta4 = HbH
0	4 deletions	Hydrops fetalis	gamma4 = Hb Barts

CIS VS TRANS MATTERS FOR REPRODUCTIVE RISK

Cis (--/alpha alpha) places both deletions on one chromosome; **trans (-alpha/-alpha)** places one on each. The companion handout highlights cis as more common in Asian ancestry and trans as more common in African ancestry. The key clinical reason to care is the probability of producing a fetus with a severe alpha-gene deletion pattern.

Molecular Genetics of Human Disease: Hemoglobinopathies



Schematic diagram of the pathogenesis of the thalassemias.
(RBCs = red blood cells)

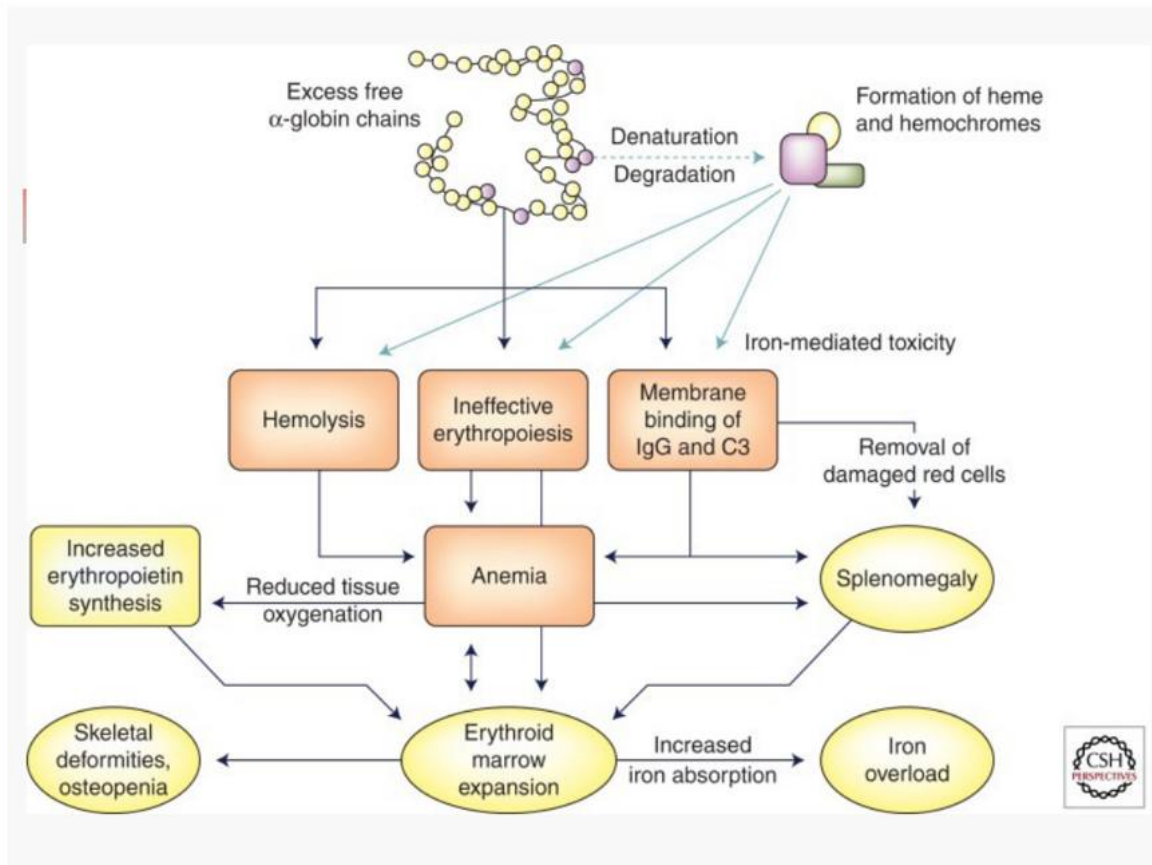
Lecture slide 32 - full slide retained. This slide gives the chain-imbalance logic. Alpha-thal leaves excess beta-like chains that can form tetramers; beta-thal leaves excess alpha chains, which are very insoluble and damage erythroid precursors and RBCs.

WHY BETA-THAL CAUSES SO MUCH INEFFECTIVE ERYTHROPOIESIS

Unpaired alpha chains are poorly soluble. They precipitate in erythroid precursors, causing oxidative/membrane injury and death in the marrow. That is why beta-thalassemia combines **ineffective erythropoiesis + hemolysis**.

LO 5 checkpoint - genetics

Can you explain why 4 alpha genes create a deletion-count phenotype, define beta+ versus beta0, and predict the excess-chain tetramer in HbH versus Hb Barts?



Lecture slide 33 - full slide retained. Follow the feedback loops: excess free alpha chains drive ineffective erythropoiesis and hemolysis; anemia drives erythropoietin and marrow expansion; ineffective erythropoiesis increases iron absorption; transfusions add still more iron.

Beta-thal trait, intermedia, major

- **Trait:** heterozygous beta⁰ or beta⁺; often asymptomatic or mild anemia; lecture emphasizes **HbA2 around 5%** and sometimes mildly increased HbF.
- **Intermedia:** moderate anemia, variable transfusion requirement, hepatosplenomegaly/hypersplenism, skeletal changes, extramedullary hematopoiesis.
- **Major:** homozygous or compound heterozygous severe disease, classically transfusion-dependent in early infancy with marrow expansion, hepatosplenomegaly, growth/puberty effects, and iron overload.

NEED TO MEMORIZE - BETA-THAL TRAIT CLUE

Microcytosis + elevated HbA2.

The slide gives HbA2 around 5% in trait.

WHY THE BONES AND FACE CHANGE

Severe chronic anemia drives massive EPO stimulation. Because erythropoiesis is ineffective, the body keeps expanding marrow space without efficiently correcting the anemia. The result is skeletal remodeling and extramedullary hematopoiesis.

Slides 38-39: conventional and curative treatment

- Regular RBC transfusion can suppress ineffective erythropoiesis and maintain oxygen delivery, but causes cumulative **iron overload**.
- **Iron chelation** is therefore inseparable from chronic transfusion therapy.
- HSCT can be curative in selected patients.
- Gene therapy aims to restore effective beta-like globin production or reactivate HbF.

CURRENT 2026 TREATMENT UPDATES

Zynteglo (betibeglogene autotemcel) is FDA-approved for adult and pediatric patients with beta-thalassemia who require regular RBC transfusions. **Casgevy** is also FDA-approved for transfusion-dependent beta-thalassemia. In January 2026, FDA approved **mitapivat (Aqvesme)** for anemia in adults with alpha- or beta-thalassemia, making the lecture's final pyruvate-kinase slide especially current.

FDA Zynteglo • FDA Casgevy • FDA mitapivat/Aqvesme update

LO 5 checkpoint - phenotype

Can you distinguish beta-thal trait/intermedia/major and explain how chain precipitation -> ineffective erythropoiesis -> marrow expansion -> iron overload?

WHY THIS SECTION EXISTS

The official Week 7 objectives explicitly require composition-analysis interpretation, but the 40-slide deck gives normal fractions and genotype descriptions rather than a full interpretation algorithm. This section is therefore **clearly labeled companion/board enrichment**, not something silently attributed to a slide.

FIRST QUESTION ON ANY HEMOGLOBIN ANALYSIS

Is HbA present? If yes, at least some normal beta chains are being produced. If no, think severe beta-chain replacement/absence (for example HbSS, beta0-thal major, or a compound variant pattern).

Pattern	Typical composition clue	Reasoning
Normal adult	Mostly A; small A2; <1% F	Normal beta production.
Sickle trait (AS)	A > S; A present	One normal beta allele makes substantial HbA.
HbSS	No HbA; S dominant; variable F/A2	No normal betaA allele.
HbSC	S + C; no A	One S allele + one C allele.
beta-thal trait	HbA remains dominant; HbA2 increased , often mild F increase	Reduced beta production shifts relative use of delta/gamma chains.
beta-thal major	Very high HbF with very low/absent HbA depending on beta+ vs beta0	Severe beta-chain deficiency.
alpha-thal trait	Adult electrophoresis may be essentially normal	All common adult Hbs require alpha chains, so proportional changes can be subtle.
HbH disease	HbH (beta4) detectable	Only one alpha gene remains.
Hydrops / Hb Barts	Hb Barts (gamma4)	No alpha chains available for fetal HbF.

EXAM TRAP

Microcytosis does not automatically mean iron deficiency. Thalassemia often produces marked microcytosis out of proportion to the degree of anemia. The companion handout also supplies the board-style Mentzer index clue: MCV/RBC count <13 favors thalassemia, while >13 favors iron deficiency.

LO 3 checkpoint

Can you look at a pattern and answer: **Is A present? Is A2 increased? Is F increased? Is a variant peak replacing A? Is the patient making any normal beta globin?**

Learning-Objective Synthesis

Official LO	What you should be able to do without notes
LO 1	Write the chain composition of HbA/A2/F, give adult fractions, explain fetal higher O ₂ affinity, locate alpha/beta genes, and explain postnatal switching.
LO 2	Define qualitative versus quantitative hemoglobinopathies and connect structural variants to missense mutations versus thalassemias to impaired chain production.
LO 3	Interpret whether HbA is present, identify high HbA ₂ , recognize S/C/F-dominant patterns, and know why alpha-thal trait may have a near-normal adult analysis.
LO 4	State HbS (beta6 Glu->Val), HbC (beta6 Glu->Lys), HbE (beta26 Glu->Lys/splicing effect), and connect genotype to hematologic/clinical behavior.
LO 5	Contrast alpha deletion-count disease with beta+/-beta0 disease, predict HbH/Barts, and distinguish beta-thal trait/intermedia/major.

One-table disease comparison

Disorder	Genetic / molecular issue	Composition clue	Signature clinical logic
HbSS	beta6 Glu->Val on both beta alleles	No A; S dominant	Polymerization -> vaso-occlusion + hemolysis
HbC	beta6 Glu->Lys	C present	Usually mild hemolysis; crystals
HbE	beta26 Glu->Lys + altered splicing	E present	Structural + mild beta-thal behavior
Alpha-thal	Usually alpha-gene deletions	Trait may look normal; HbH/Barts in severe forms	Severity tracks deleted-gene count
Beta-thal trait	One beta+ or beta0 allele	HbA2 increased	Microcytosis, mild/no symptoms
Beta-thal major	Severe two-allele beta deficiency	High F; low/absent A	Ineffective erythropoiesis + transfusion dependence + iron overload

LAST 60 SECONDS

A = alpha2 beta2. A2 = alpha2 delta2. F = alpha2 gamma2.

S = beta6 Glu->Val; C = beta6 Glu->Lys; E = beta26 Glu->Lys + splicing effect.

Alpha-thal = delete alpha genes; 3 deletions -> beta4 HbH, 4 deletions -> gamma4 Barts.

Beta-thal = beta+ reduced, beta0 absent; trait -> high A2; major -> high F + severe ineffective erythropoiesis.

Source hierarchy: Raval/Kutlar lecture slides are the primary slide-order source. The uploaded Lecture 9 companion is used only where it adds an official-LO item that the slide deck does not develop (especially HbC/HbE and composition-analysis approach), and those additions are labeled. Current FDA treatment updates are explicitly separated as external/current enrichment. Official Week 7 objectives determine the emphasis.